Exploring Risk Gradients

Andy Davidson

2025-07-08

This document contains an attempt to explore how the relative contribution of NCEs to the total predator effect varies across a risk gradient. This exploration is an attempt to flesh out the theory paper and help distinguish it from a rehash of Scott's 2013 paper in Oecologia and ideas in the 2020 Ecology paper.

This is similar to Fig. 4 in the grant proposal, which uses predator density as a proxy for risk.

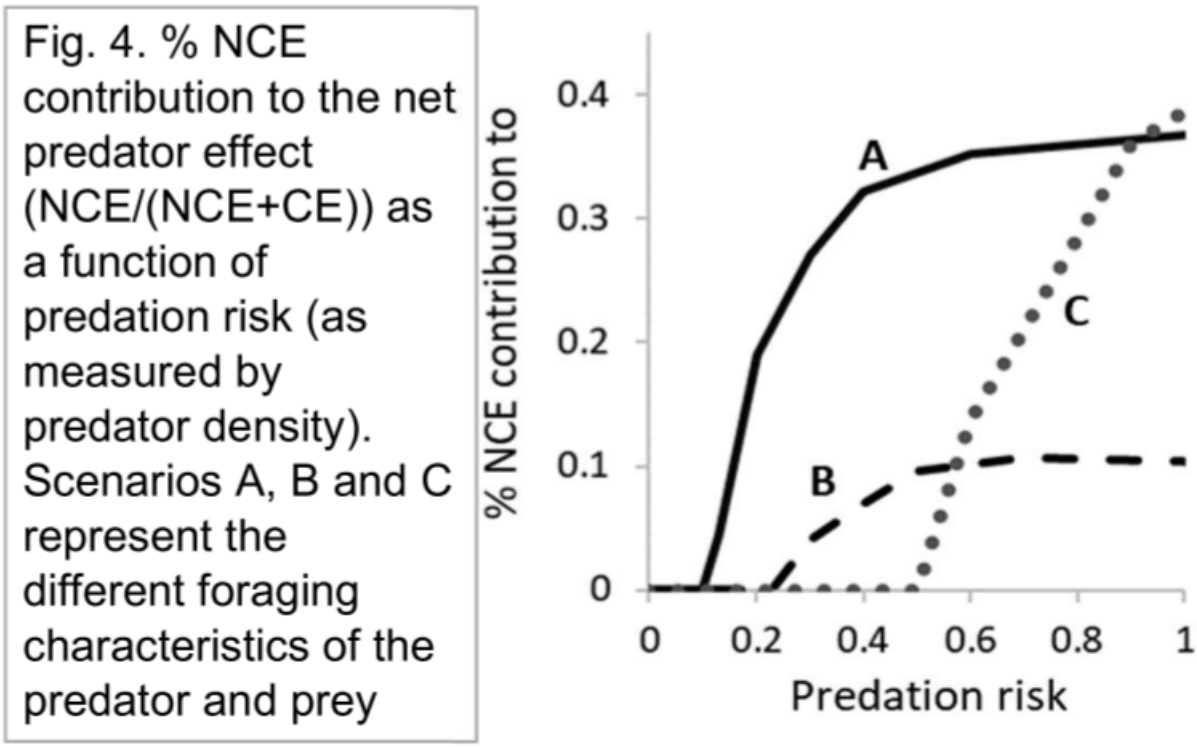

Figure 1: Fig. 4 from Grant Proposal

Notably, the curves generally reach a saturation point, where the contribution of NCEs to the total predator effect levels off with increasing risk. One goal of this document is determining why the percentage of the total predator effect composed of the NCE generally levels off given increasing risk.

Exploring the Mechanisms

We will first recreate curve A in the above figure, but on a finer scale. Specifically, we will look at the underlying IREF curves and their shapes to determine why this relationship saturates with increasing risk. In the following figures, the only parameter being changed is predator density, with all other parameters (predator displacement rate, prey handling time, etc.) are held constant.

This uses the following equations for the mortality and growth curves, respectively, as in the grant:

$$Z = 2rN * sqrt(v^2 + s^2)$$

where Z is the encounter rate between the predator and prey, s is prey speed (i.e., displacement rate), v is predator speed, N is predator density, and r is the radius within which the predator can perceive prey; and:

$$f(R) = \frac{sR}{1 + shR}$$

where $f(R)$ is the growth curve, s is prey displacement rate, h is handling time, and R is resource density (held constant at $R = 1$).

```r
# Setting the range of prey displacement rate
trait.vec <- seq(0, 1, by = 0.01)
speed.vec <- 1 - trait.vec

# These formulas are simplified by setting some of the
# values as constants r is always assumed to be 1, while k,
# s, and v are manipulated values
pred.curve <- function(k, v, s) {
    k * sqrt(v^2 + s^2)
}

# R is always assumed to be 1, while s and h are
# manipulated values
growth.curve <- function(s, R, h) {
    (R * s)/(1 + s * R * h)
}

# Range of values used by Scott for Fig. 4 in the grant
# proposal
k_vec <- c(0, 0.1, 0.13, 0.2, 0.3, 0.4, 0.6, 1, 1.5, 2, 3, 5)

h_val <- 8
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
pred_constant_val <- 0.98058

# Generate growth and mortality curves for each value of
# predator density (k_vec)
fig4.df <- as.data.frame(mat.or.vec(length(trait.vec) * length(k_vec),
    7))
colnames(fig4.df) <- c("trait", "speed", "growth_curve", "mort_curve",
    "fitness", "pct_NCE", "pred_dens")

row <- 0

for (i in 1:length(k_vec)) {
    for (j in 1:length(trait.vec)) {
        row <- row + 1
        fig4.df[row, 1] <- trait.vec[j]
        fig4.df[row, 2] <- speed.vec[j]
        fig4.df[row, 3] <- cost_constant_val * growth.curve(s = fig4.df[row,
            2], R = R_val, h = h_val)
        fig4.df[row, 4] <- pred_constant_val * pred.curve(k = k_vec[i],
            v = v_val, s = fig4.df[row, 2])
        fig4.df[row, 5] <- fig4.df[row, 3] - fig4.df[row, 4]
        fig4.df[row, 6] <- (1 - fig4.df[row, 3])/((1 - fig4.df[row,
            3]) + fig4.df[row, 4])
        fig4.df[row, 7] <- k_vec[i]
    }
}
```

First, we will look at the NCE's relative contribution to the total predator effect at the optimal RITR, as a function of predator density.

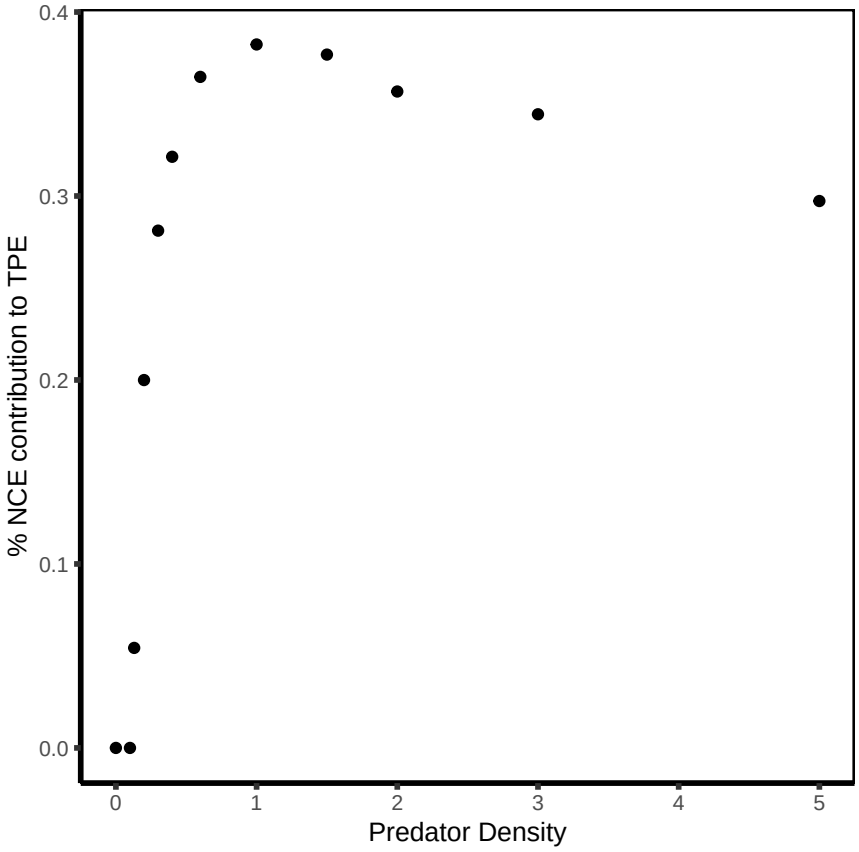

Fig. 2: The proportion of the total predator effect (TPE) made up of NCEs, as a function of increasing predator density (i.e., predation risk).

This shows an initially saturating relationship as seen in the grant figure (Fig. 1), but I've gone beyond the range of densities used there. This shows that it is actually a hump-shaped relationship that begins to decline with increasing density.

Next, we'll look under the hood at the underlying IREF curves and the optimal RITRs, as a function of predator density.

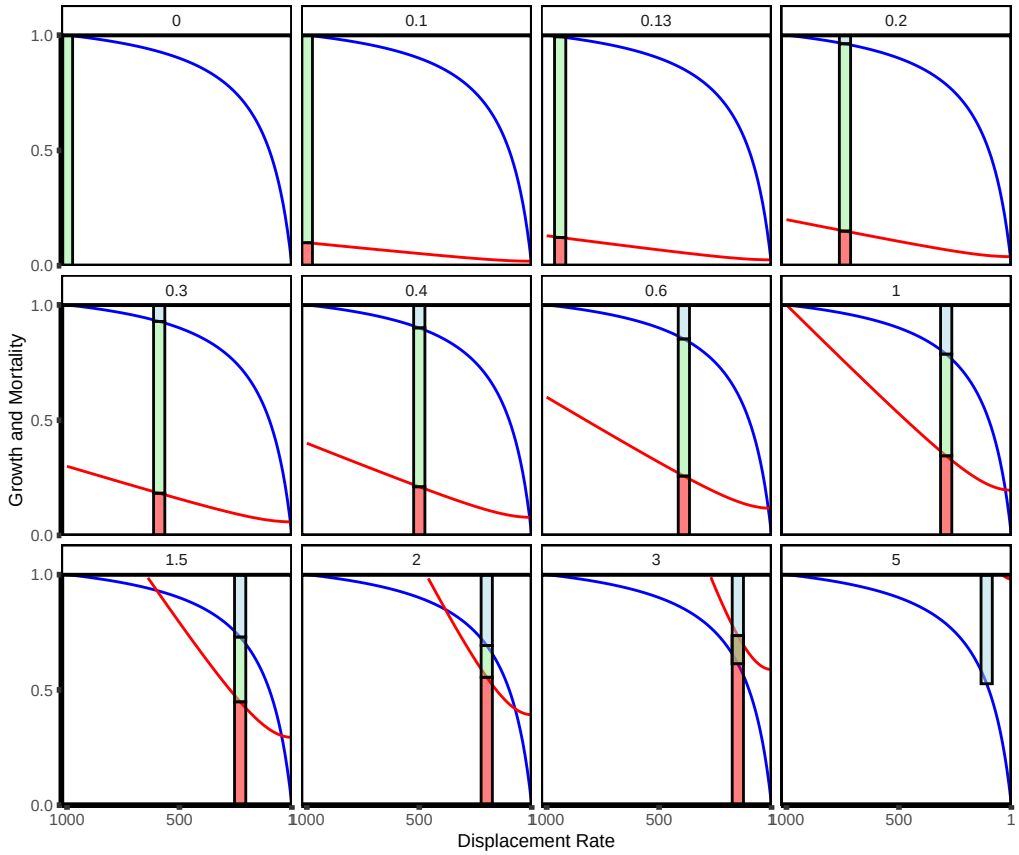

Fig. 3: Growth and mortality curves (blue and red, respectively) as a function of predator density (noted in the header above the plot). The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength. The bars eventually "fall apart" at very high predator densities ($k > 3$) because fitness goes negative.

The reason that the NCE's contribution to the total predator effect tends to exhibit a hump-shaped relationship with predator density depends on the relative costs and benefits of the RITR.

Increasing risk generally shifts the optimal RITR towards the right (i.e., towards less prey activity), but there is a limit to how strongly the prey can respond. This is determined by the point at which the slope of the growth curve exceeds the slope of the mortality curve.

The proportion of the total predator effect composed of NCEs eventually declines, because as predator density increases, CEs continue to grow while NCEs are bounded by a prey displacement rate of zero.

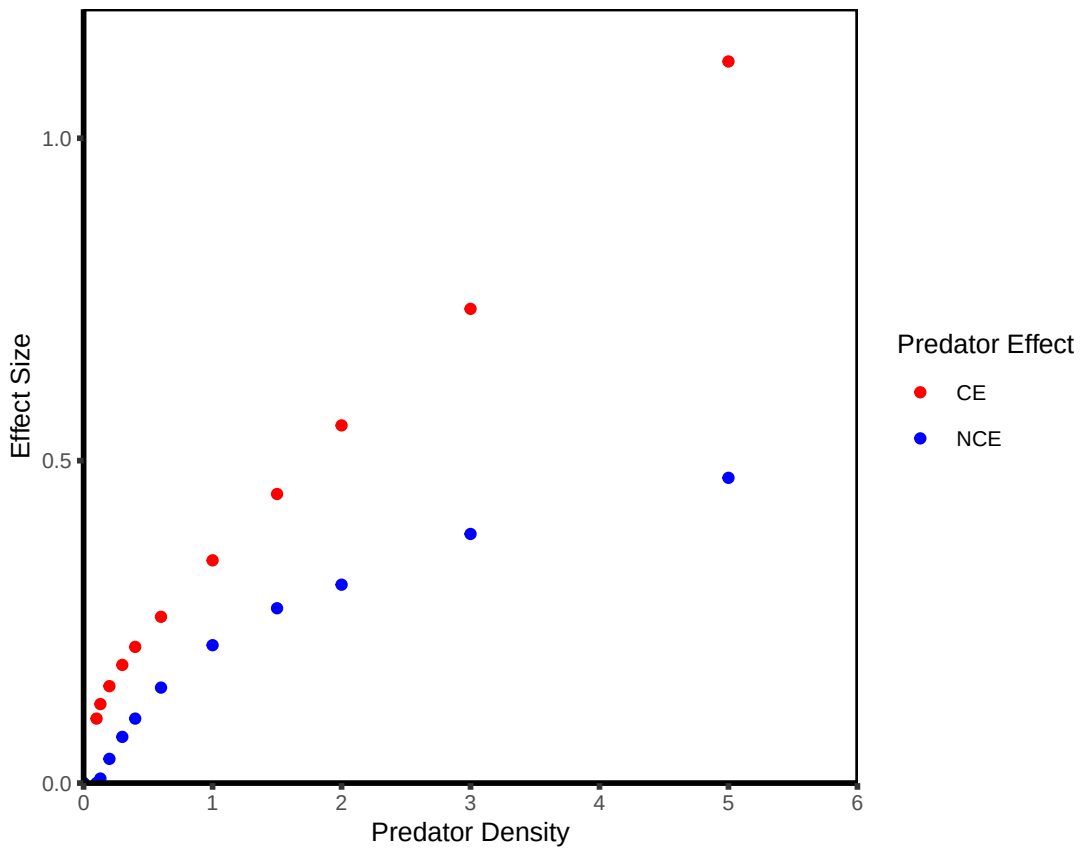

Fig. 4: The effect size of NCEs (blue points) and CEs (red points) as a function of predator density at the optimal RITR.

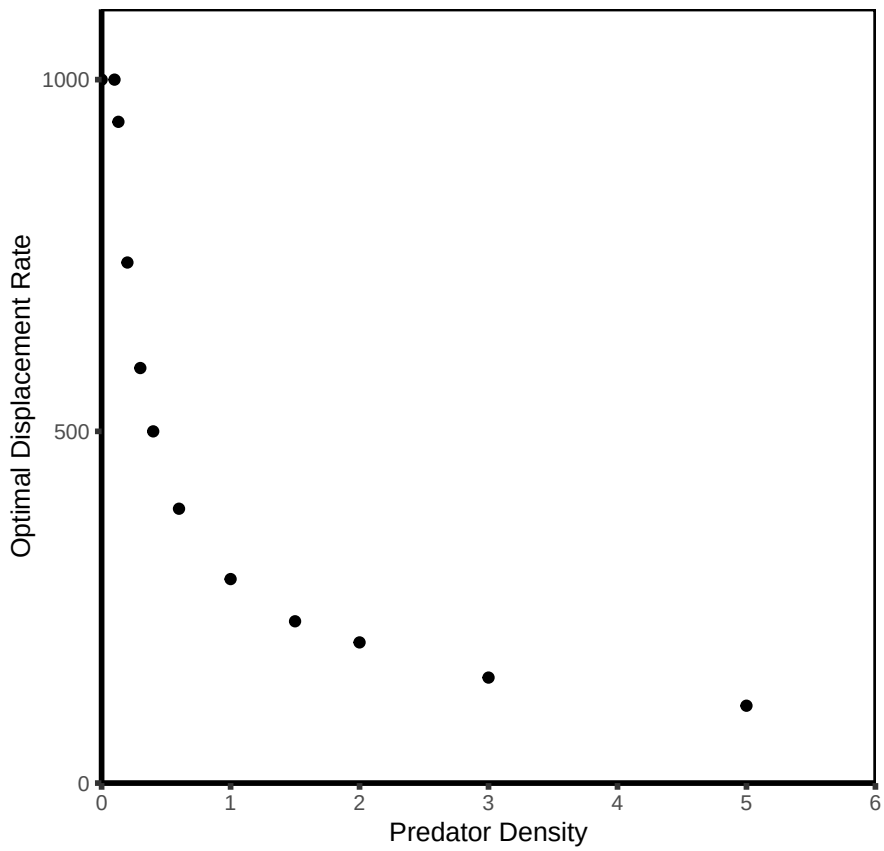

Fig. 5: The optimal RITR (displacement rate) as a function of predator density.

These findings are probably general in cases where the growth curve is concave-down, while predation risk declines in a linear or saturating way. I would also suggest that the "relative slopes" approach to determining the magnitude of the prey RITR is probably a good rule of thumb, as Scott suggests in Peacor et al. 2013. I.e., prey should only reduce their foraging rates if the slope of the growth curve is greater than the slope of the mortality curve (i.e., the benefits of reducing foraging rates are growing faster than their costs).

What if the Predator has a Type II Functional Response?

Next, I want to explore whether the above finding would change if the predator had a Type II functional response, as is typical for most predators. This may help to establish whether this finding is "general" to the majority of predator-prey systems.

I expect that this will change the mortality curve primarily when prey displacement rate (s) is high. At high prey displacement rates, predator handling time will cause the mortality curve to be lower than with a Type I functional response. The optimal RITR will differ most from the Type I functional response examples above in this "saturating" region at high prey displacement rates, where it may take a larger reduction in prey displacement rate for the costs to growth to outweigh the benefits in terms of reduction of mortality.

To explore this, I have taken the encounter rate equation above and nested it inside of a Type II functional response, assuming that encounter rate and attack rate are roughly equal ($Z \approx a$; i.e., an encounter generally leads to an attack in the absence of handling time):

$$f(C) = \frac{ZC}{1 + Zh_{pred}C}$$

where $f(C)$ is the mortality curve, Z is the predator's encounter rate with its prey, h_{pred} is the predator's handling time, and C is prey density.

We'll start with an (arbitrary) predator handling time (h_{pred}) of 5, while holding all other parameters constant (including prey density, but see below), except for the scaling constant applied to predator foraging rates. I adjusted this value from ~1 to 5, because the introduction of handling time to the equation strongly reduces predator feeding rates.

```
# First, write a new Type II mortality curve This function
# will take the above encounter rate formula and apply
# handling time to it
typeII.curve <- function(a_pred, h_pred) {
    a_pred/(1 + a_pred * h_pred)
}

# Range of values used by Scott for Fig. 4 in the grant
# proposal
k_vec <- c(0, 0.1, 0.13, 0.2, 0.3, 0.4, 0.6, 1, 2, 3, 5)

h_val <- 8
h_pred_val <- 5  # Start at a predator handling time value of 5
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
pred_constant_val <- 5

# Generate growth and mortality curves for each value of
# predator density (k_vec)
typeII.df <- as.data.frame(mat.or.vec(length(trait.vec) * length(k_vec),
    7))
colnames(typeII.df) <- c("trait", "speed", "growth_curve", "mort_curve",
    "fitness", "pct_NCE", "pred_dens")

row <- 0

for (i in 1:length(k_vec)) {
```

```
    for (j in 1:length(trait.vec)) {
        row <- row + 1
        typeII.df[row, 1] <- trait.vec[j]
        typeII.df[row, 2] <- speed.vec[j]
        typeII.df[row, 3] <- cost_constant_val * growth.curve(s = typeII.df[row,
            2], R = R_val, h = h_val)
        typeII.df[row, 4] <- pred_constant_val * typeII.curve(a_pred = pred.curve(k = k_vec[i],
            v = v_val, s = typeII.df[row, 2]), h_pred = h_pred_val)
        typeII.df[row, 5] <- typeII.df[row, 3] - typeII.df[row,
            4]
        typeII.df[row, 6] <- (1 - typeII.df[row, 3])/((1 - typeII.df[row,
            3]) + typeII.df[row, 4])
        typeII.df[row, 7] <- k_vec[i]
    }
}
```

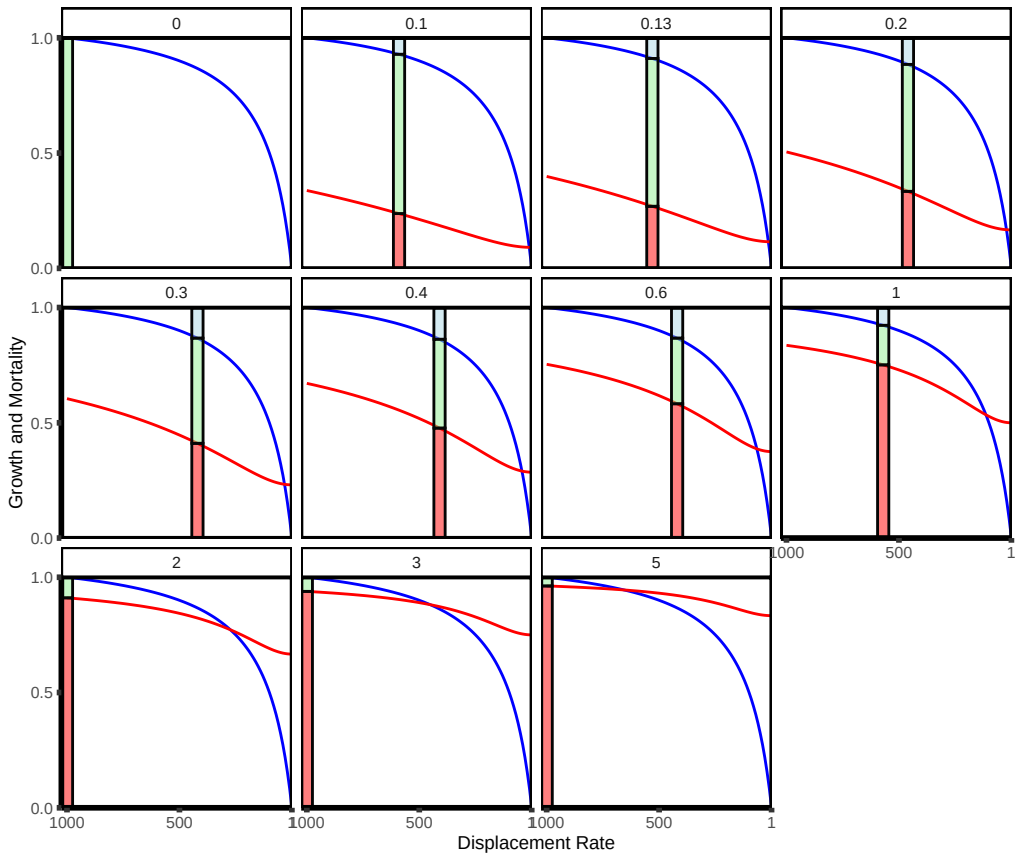

Fig. 6: Growth and mortality curves (blue and red, respectively) as a function of predator density (noted in the header above the plot), when predators exhibit a Type II Functional Response. The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength.

The behavior of the fitness curves is similar looking, at least until predator density exceeds one, where the optimal RITR begins to shift back to the left. Biologically, this is likely because predation risk is so high that there is no fitness gain to be had from reducing foraging activity.

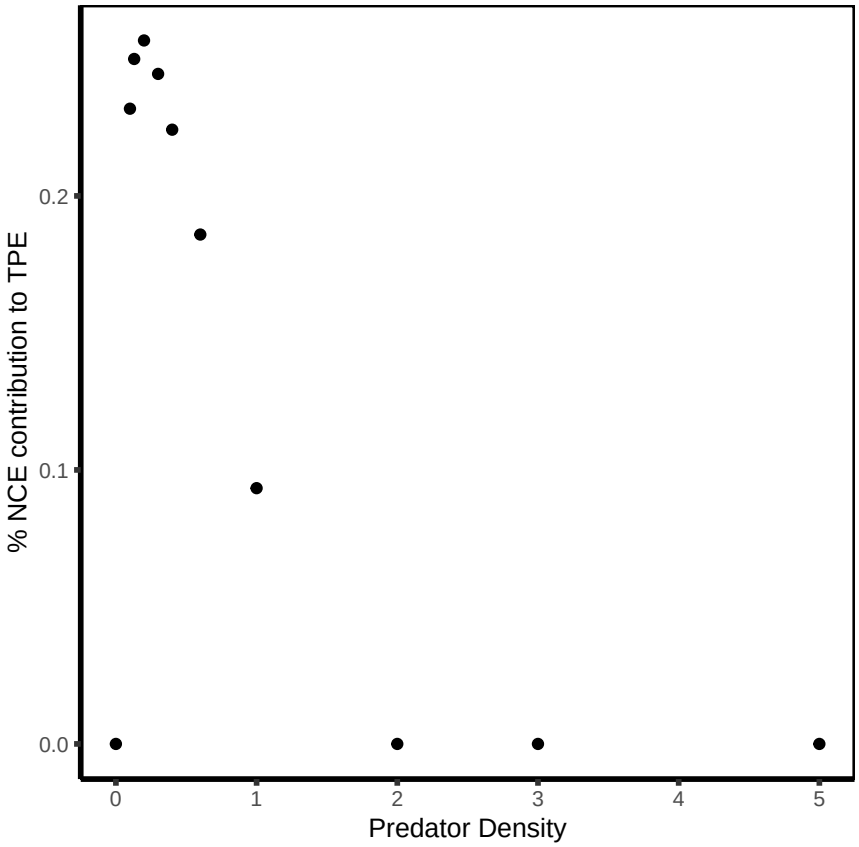

Fig. 7: The proportion of the total predator effect (TPE) made up of NCEs, as a function of increasing predator density (i.e., predation risk), when predators exhibit a Type II Functional Response.

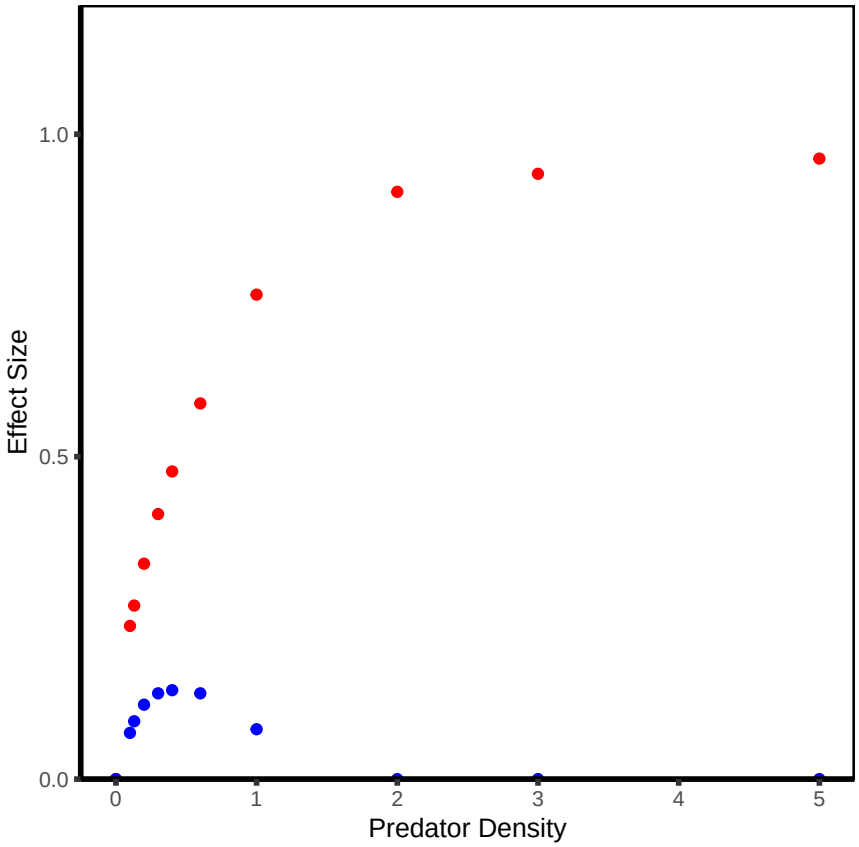

Fig. 8: The effect size of NCEs (blue points) and CEs (red points) as a function of predator density at the optimal RITR, when predators exhibit a Type II Functional Response.

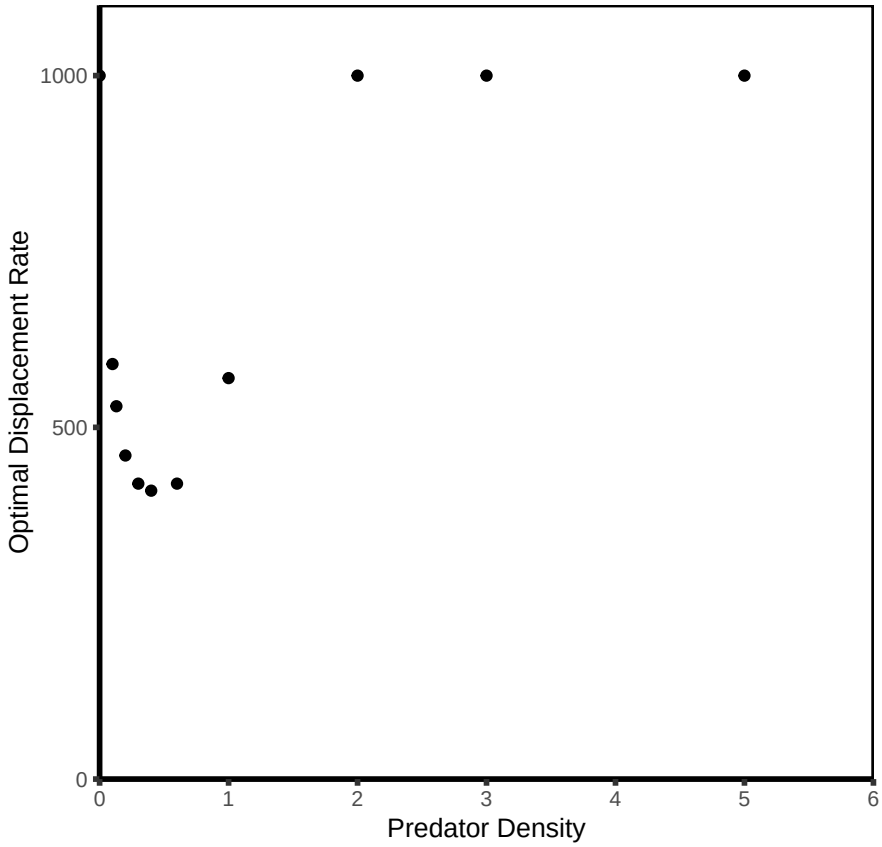

Fig. 9: The optimal RITR (displacement rate) as a function of predator density, when predators exhibit a Type II Functional Response.

The relative contribution of the NCE to the total predator effect reaches its peak and declines much sooner. This is because as predator density increases, prey switch from reducing their foraging rates to not responding to predators.

Next, we'll repeat the above scenarios but instead of manipulating predator density, we will manipulate prey density. This will help us examine how our predictions change with increasing predator satiation.

We will hold predator density constant at $k = 0.6$, while varying prey density, C, between 0.1 and 2.5.

```
typeII_full.curve <- function(a_pred, h_pred, C) {
    (a_pred * C)/(1 + a_pred * h_pred * C)
}

k_val <- 0.6  # Range of values used by Scott for Fig. 4 in the grant proposal
h_val <- 8
h_pred_val <- 5  # Start at a predator handling time value of 5
R_val <- 1
C_vec <- c(0.1, 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5)
cost_constant_val <- 9
v_val <- 0.2
pred_constant_val <- 5

# Generate growth and mortality curves for each value of
# prey density (C_vec)
typeII_full.df <- as.data.frame(mat.or.vec(length(trait.vec) *
    length(C_vec), 8))
colnames(typeII_full.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "pct_NCE", "pred_dens", "prey_dens")

row <- 0

for (i in 1:length(C_vec)) {
    for (j in 1:length(trait.vec)) {
        row <- row + 1
        typeII_full.df[row, 1] <- trait.vec[j]
        typeII_full.df[row, 2] <- speed.vec[j]
        typeII_full.df[row, 3] <- cost_constant_val * growth.curve(s = typeII_full.df[row,
            2], R = R_val, h = h_val)
        typeII_full.df[row, 4] <- pred_constant_val * typeII_full.curve(a_pred = pred.curve(k = k_val,
            v = v_val, s = typeII_full.df[row, 2]), h_pred = h_pred_val,
            C = C_vec[i])
        typeII_full.df[row, 5] <- typeII_full.df[row, 3] - typeII_full.df[row,
            4]
        typeII_full.df[row, 6] <- (1 - typeII_full.df[row, 3])/((1 -
            typeII_full.df[row, 3]) + typeII_full.df[row, 4])
        typeII_full.df[row, 7] <- k_val
        typeII_full.df[row, 8] <- C_vec[i]
    }
}
```

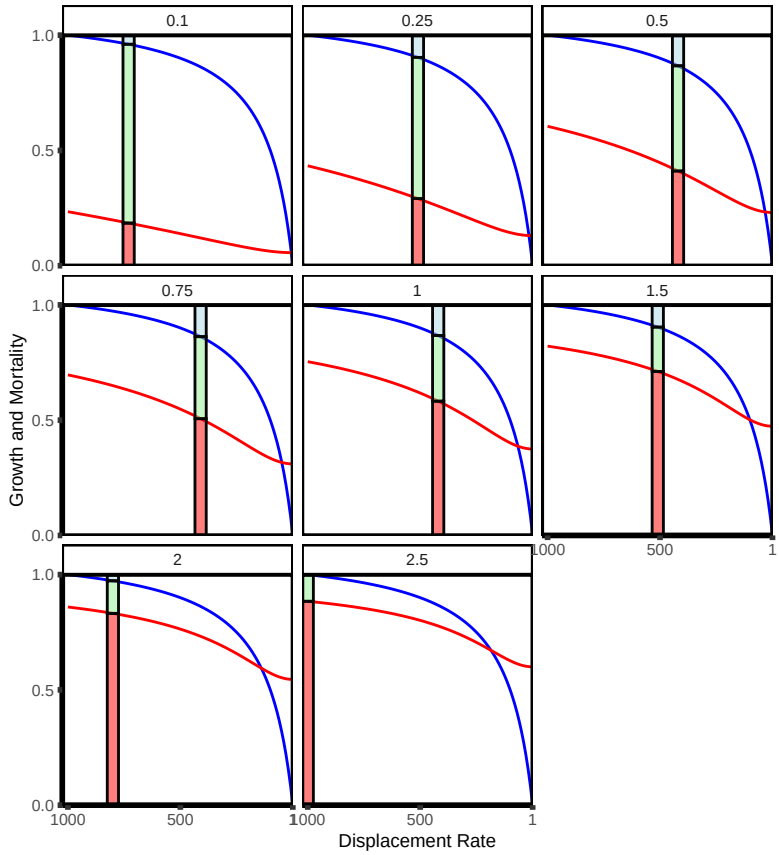

Fig. 10: Growth and mortality curves (blue and red, respectively) as a function of prey density (noted in the header above the plot), when predators exhibit a Type II Functional Response. The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength.

Increasing prey density causes a stronger reduction in mortality at higher prey displacement rates, which is expected because the effect of handling time should become stronger when encounters with prey are frequent (which can be due to displacement rate or prey density).

As prey density increases, the optimal RITR at first shifts to the right, then back to the left. Mathematically, this is probably because as prey density increases, the saturating effect of predator handling time makes it "safer" for prey to be out in the open foraging actively (i.e., because if they do encounter predators, they might not be hungry). But biologically speaking, I don't know how plausible this is. Do individual prey "know" how dense their populations are, and do they have any sense of when/if predators will be satiated? These are population level traits, instead of individual level traits.

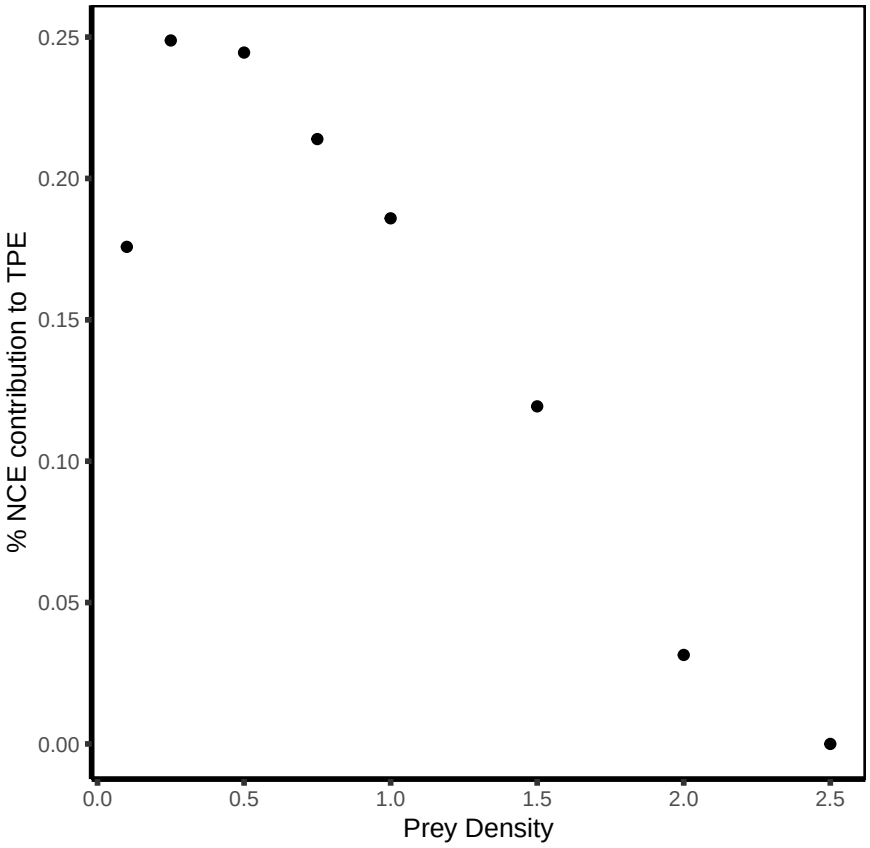

Fig. 11: The proportion of the total predator effect (TPE) made up of NCEs, as a function of increasing prey density, when predators exhibit a Type II Functional Response.

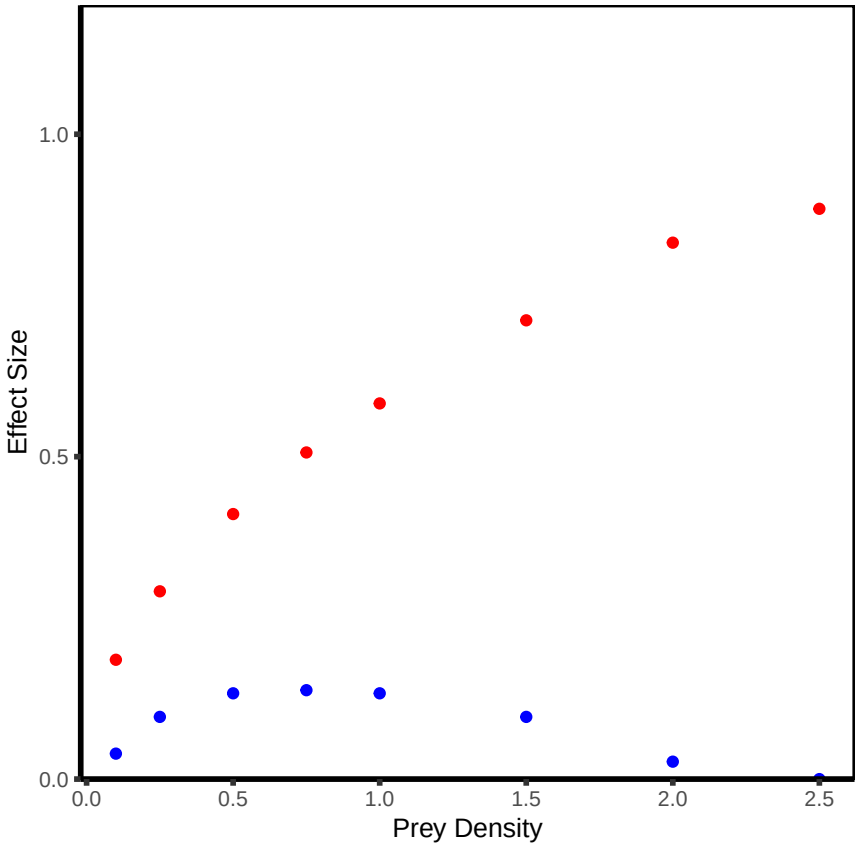

Fig. 12: The effect size of NCEs (blue points) and CEs (red points) as a function of prey density at the optimal RITR, when predators exhibit a Type II Functional Response.

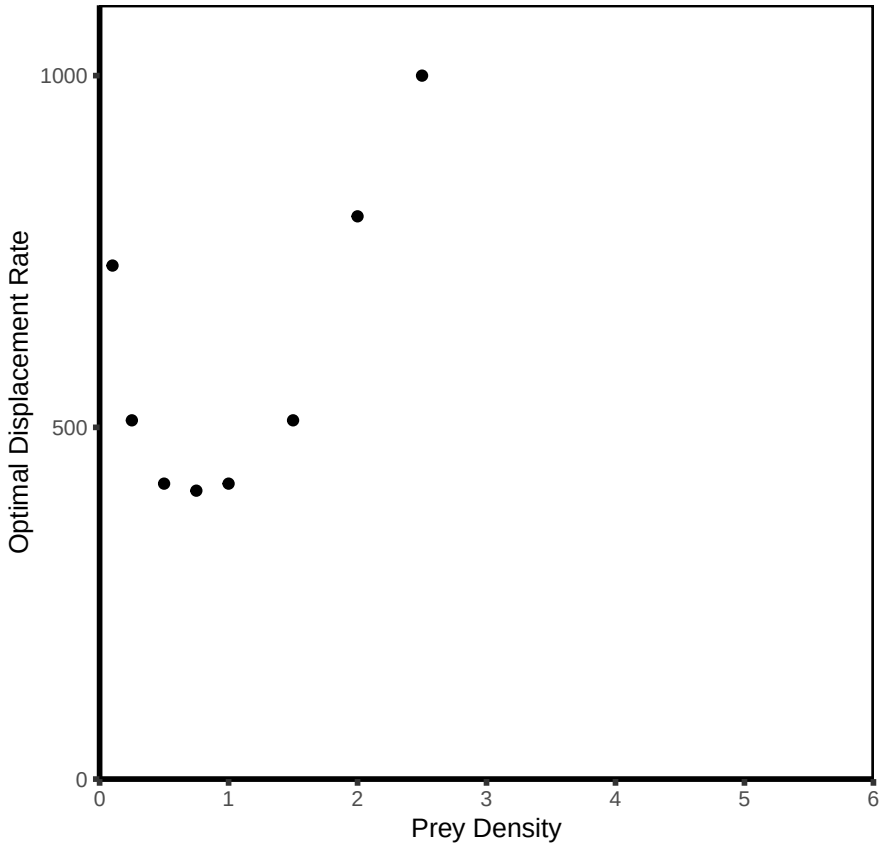

Fig. 13: The optimal RITR (displacement rate) as a function of prey density, when predators exhibit a Type II Functional Response.